

University of Waterloo
Co-operative Work Terms

Leo You
20998391

2B Mechatronics Engineering, Honours, Co-operative Program

Work Term	Employer	Evaluation
Jan - Apr 2023	MARTINREA International Inc Hydroform Solutions Brampton Ontario Canada Engineering Student	OUTSTANDING
Sep - Dec 2023	Arctic AI Canada Division Toronto Ontario Canada Junior Developer	OUTSTANDING
May - Aug 2024	EnStream LP Divisional Office Toronto Ontario Canada Software and Data Engineering	OUTSTANDING

Planned Future Work Term(s)

Jan - Apr 2025
Sep - Dec 2025
May - Aug 2026

Disclaimer: This evaluation does not constitute an employment endorsement or recommendation. Employer evaluations of student contributions and achievements during the work term are conducted as part of the University of Waterloo's Co-operative (Co-op) Education model. Like academic grades, overall evaluations are part of the assessment of a student's progress in the co-op portion of their degree studies. These assessments are completed using criteria set out by the University, not the employer, and do not reflect the employer's criteria or assessment metrics.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, MATLAB, VHDL, Bash, SQL, HTML/CSS
Frameworks and Libraries: TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, React, Next.js, Express.js, Node.js, Django, MongoDB, Passport.js, Knex.js, OAuth 2.0, Firebase, Tailwind CSS
Tools and Technologies: Linux/Unix, Git, SVN, Docker, GCP, AutoCAD, SolidWorks
Hobbies: Painting, Literature, Kickboxing, Badminton, Hiking

EXPERIENCE

Software and Data Engineering Intern

May 2024 – Aug 2024

*EnStream LP**Toronto, ON*

- Developed products for identifying and detecting clusters of telecom fraud with **Python** and **SQL** for Canadian telecommunications, banking, and insurance partners.
- Analyzed distributions and time-series indicators curated from customer, device, and activity data with **Matplotlib** and **Pandas** to extract consumer behaviour patterns, develop a trust-scoring function, and curate high-value indicators for supervised machine learning.
- Designed a product demo website with **Next.js**, **MySQL**, and **Docker** to demonstrate multifactor authentication based on verifying the request origin with the telephone number on file.

Software Developer Intern

Sep. 2023 – Dec 2023

*Arctic AI**Toronto, ON*

- Consulted on and led healthcare administration application development for SaaS startup, reducing labor intensity by **95%** and generating **\$28,000** in revenue.
- Designed REST APIs and tools for scalable content and data storage, email campaigns, and user management features secured by JWT cookie-based authentication using a **MySQL**, **Express.js**, **TypeScript** **React**, and **TypeScript** stack and deployed with **Docker** on **GCP**.
- Developed internal SDK and codebase for database management and migration, secure authentication, and microservice standards to streamline application development.

Software Developer Co-op

Jan. 2023 – April 2023

*Martinrea Hydroform Solutions**Brampton, ON*

- Oversaw production line camera vision project that determines correct machine-line part presence and orientation with **OpenCV** image processing, reducing production downtime by **1.14%** and saving **\$60,000** per year.
- Scripted object detection and mask generation programs using **OpenCV** and **NumPy**, detecting loading accuracy for 47 distinct automotive components.

PROJECTS

Rotary Encoder-based Gauge | *Arduino, SolidWorks, 3D-Printing*

- Created and calibrated a gauge that converts encoder ticks to linear distances with an accuracy of ± 3.5 mm, designing and prototyping device chassis, transmission rack, and gear pinion to completion within 3 days.

Journal++ | *TensorFlow, JavaScript, Python, React.js, Django, Git*

- Developed a full-stack web application serving a **NLP**-powered smart journal that organizes thoughts based on sentiment and topic.
- Trained a sentiment classification machine learning model with **TensorFlow** achieving **93%** accuracy.

Daily Automatic Date Scribe | *SolidWorks, 3D Printing, Rapid Prototyping*

- Developed an autonomous LEGO EV3 robot that draws and updates the date on a whiteboard, integrating code for path logic, movement, and calibration and prototyping custom mount to stabilize marker movement.

Spotify to YouTube Playlist Migration | *JavaScript, OAuth 2.0, Firebase*

- Developed a Google Chrome extension with Google and Spotify **OAuth 2.0** and developer APIs to authenticate and migrate a user's Spotify playlist to YouTube.

EDUCATION

University of Waterloo

Waterloo, ON

*Candidate for Honours B.A.Sc. in Mechatronics Engineering**April 2027***GPA:** 93.44%**Relevant Coursework:** Data Structures and Algorithms, Real-Time Operating Systems, Statistics, Linear Algebra

UNIVERSITY OF WATERLOO

UNOFFICIAL GRADE REPORT

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2B Mechatronics Engineering, Honours, Co-operative Program

Fall 2024

MTE	241	Comp Struct & Real-Time Sys	
MTE	252	Linear Systems & Signals	
MTE	200B	Seminar	
MTE	203	Advanced Calculus	
MTE	204	Numerical Methods	
WKRPT	200	Work-term Report	*
		*Degree Requirement, Not in Average	
MTE	220	Sensors & Instrumentation	
Term Average:		N/A	Decision:

Spring 2024

PD	11	Technical Report Writing	CR
COOP	3	Co-operative Work Term	CR
Term Average:		N/A	Decision:

Winter 2024

SYDE	182	Physics 2: Dynamics	94
MTE	202	Ordinary Differential Equation	93
WKRPT	100	Work-term Report	89*
		*Degree Requirement, Not in Average	
MTE	219	Mechanics of Deformable Solids	95
MTE	200A	Seminar	
MTE	201	Exp Measure & Stat Analysis	89
MTE	262	Microprocessors & Digi Logic	97
STV	305	Tech, Society & Modern City	88
MATH	135	Algebra (Honours)	81*
		*Not Required, Not in Average	
CS	136L	Tools & Tech for Software Dev	CR*
		*Not Required, Not in Average	
Term Average:		92.67	Decision: Excellent Standing

Fall 2023

COOP	2	Co-operative Work Term	CR
PD	20	Strategies for Career Success	CR
Term Average:		N/A	Decision:

Spring 2023

MTE	111	Material Struct & Props	96
MTE	140	Algorithms & Data Structures	93
MTE	120	Circuits	94
MATH	118	Calculus 2 (Eng)	89
MTE	119	Statics	97
MTE	100B	Seminar	
Term Average:		93.82	Decision: Excellent Standing

Winter 2023

PD	19	Tactics for Workplace Success	CR
COOP	1	Co-operative Work Term	CR
Term Average:		N/A	Decision:

Fall 2022

GENE	119	Problems Seminar	
MATH	115	Linear Algebra (Eng)	96
MTHHEL	99	First-Year Math Readiness	CR
		First year readiness	
CHE	102	Chemistry for Engineers	96

MATH	116	Calculus 1 (Eng)	92
MTE	100	Mechatronics Engineering	91
MTE	121	Digital Computation	96
Term Average: 93.91		Decision: Excellent Standing	